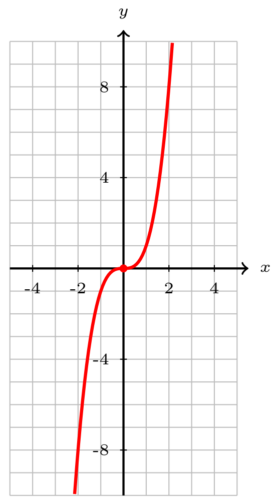
Unit 6.4: Cubic & Cube Root Functions Name: **ANSWER KEY**

Day 1 Practice: Parent Functions, Intercepts, & Key Features Period: ________

Directions: For each function, complete the table, graph, and identify the key features.

1. $f(x) = x^3$

x	y
-2	**-8**
-1	**-1**
0	**0**
1	**1**
2	**8**

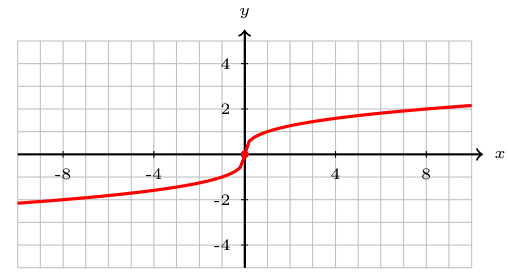


Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

x-int: $(0, 0)$ y-int: $(0, 0)$

2. $f(x) = \sqrt[3]{x}$

x	y
-8	**-2**
-1	**-1**
0	**0**
1	**1**
8	**2**



Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

x-int: $(0, 0)$ y-int: $(0, 0)$

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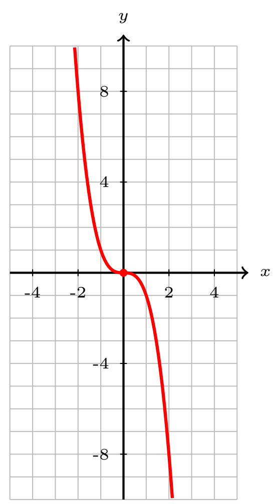
3. $f(x) = -x^3$

x	y
-2	**8**
-1	**1**
0	**0**
1	**-1**
2	**-8**

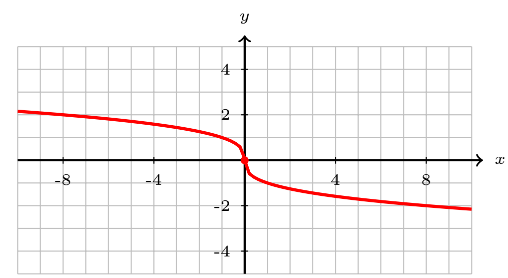


Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

x-int: $(0, 0)$ y-int: $(0, 0)$

4. $f(x) = -\sqrt[3]{x}$

x	y
-8	**2**
-1	**1**
0	**0**
1	**-1**
8	**-2**



Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

x-int: $(0, 0)$ y-int: $(0, 0)$

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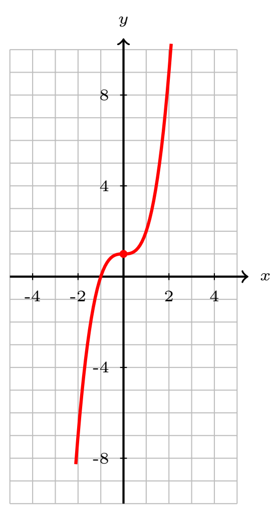
5. $f(x) = x^3 + 1$

x	y
-2	**-7**
-1	**0**
0	**1**
1	**2**
2	**9**

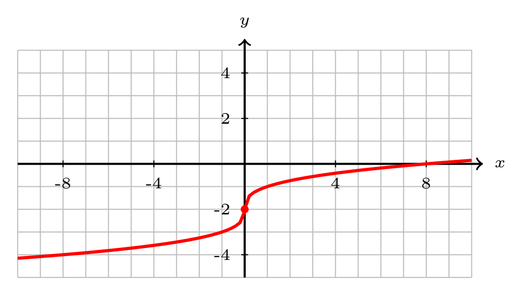


Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

x-int: $(-1, 0)$ y-int: $(0, 1)$

6. $f(x) = \sqrt[3]{x} - 2$

x	y
-8	**-4**
-1	**-3**
0	**-2**
1	**-1**
8	**0**

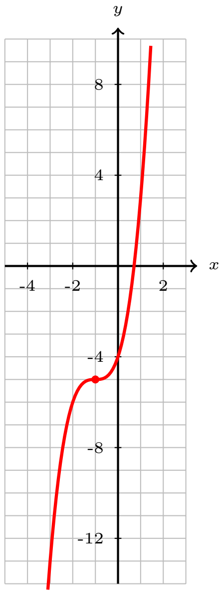


Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

x-int: $(8, 0)$ y-int: $(0, -2)$

7. $f(x) = (x+1)^3 - 5$

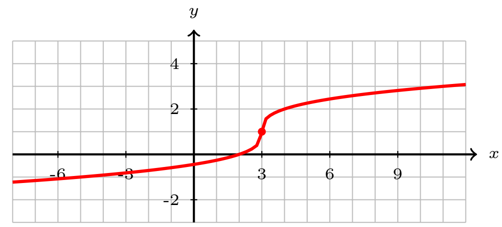
x	y
-3	**-13**
-2	**-6**
-1	**-5**
0	**-4**
1	**3**



Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$
x-int: ≈ 0.71 y-int: $(0, -4)$

8. $f(x) = \sqrt[3]{x-3} + 1$

x	y
-5	**-1**
2	**0**
3	**1**
4	**2**
11	**3**



Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$
x-int: $(2, 0)$ y-int: ≈ -0.44

Intercept Practice (No Graphing Required)

Find the x- and y-intercepts for each function. Show your work!

9. $f(x) = (x-3)^3 + 8$

y-int: Let $x = 0$
$y = (0-3)^3 + 8 = -27 + 8 = -19$

x-int: Let $y = 0$
$(x-3)^3 = -8 \Rightarrow x - 3 = -2 \Rightarrow x = 1$

x-int: **(1, 0)** y-int: **(0, -19)**

10. $f(x) = \sqrt[3]{x+27} - 3$

y-int: Let $x = 0$
$y = \sqrt[3]{27} - 3 = 3 - 3 = 0$

x-int: Let $y = 0$
$\sqrt[3]{x+27} = 3 \Rightarrow x + 27 = 27 \Rightarrow x = 0$

x-int: **(0, 0)** y-int: **(0, 0)**

11. $f(x) = (x-1)^3 - 4$

y-int: Let $x = 0$
$y = (-1)^3 - 4 = -1 - 4 = -5$

x-int: Let $y = 0$
$(x-1)^3 = 4 \Rightarrow x = 1 + \sqrt[3]{4} \approx 2.59$

x-int: ≈ 2.59 y-int: **(0, -5)**

12. $f(x) = \sqrt[3]{x+5} - 2$

y-int: Let $x = 0$
$y = \sqrt[3]{5} - 2 \approx -0.29$

x-int: Let $y = 0$
$\sqrt[3]{x+5} = 2 \Rightarrow x + 5 = 8 \Rightarrow x = 3$

x-int: **(3, 0)** y-int: ≈ -0.29

13. $f(x) = -2(x+3)^3 + 6$

y-int: Let $x = 0$
$y = -2(27) + 6 = -54 + 6 = -48$

x-int: Let $y = 0$
$-2(x+3)^3 = -6 \Rightarrow (x+3)^3 = 3 \Rightarrow x = \sqrt[3]{3} - 3$

x-int: ≈ -1.56 y-int: **(0, -48)**

14. $f(x) = 2\sqrt[3]{x-4} - 3$

y-int: Let $x = 0$
$y = 2\sqrt[3]{-4} - 3 \approx 2(-1.59) - 3 = -6.18$

x-int: Let $y = 0$
$2\sqrt[3]{x-4} = 3 \Rightarrow x - 4 = (1.5)^3 \Rightarrow x = 4 + 3.375$

x-int: **(7.375, 0)** y-int: ≈ -6.18